

INVOLVEMENT OF BRAIN IN EMOTIONS

Emotion structures in brain

Positron Emission Tomography (PET) scanning and functional MRI studies have shown that some emotions are more likely to be associated with different regions of limbic system activity than other emotions.

1. **Happiness** activates several areas of the brain, including the right frontal cortex, the left amygdala, and the left insula. This activity involves connections between awareness (frontal cortex and insula) and the “feeling centre” (amygdala) of the brain.
2. **Fear** activates the bilateral amygdala, the hypothalamus and areas of the left frontal cortex. This involves some thinking (frontal cortex), a “gut” feeling (amygdala), and a sense of urgency typically associated with survival (the hypothalamus.)
3. **Sadness** is associated with increased activity of the right occipital lobe, the left insula, the left thalamus, the amygdala and the hippocampus. The hippocampus is strongly linked with memory, and it makes sense that awareness of certain memories is associated with feeling sad. Sadness has been studied more than the other emotions because depression may last for a long time; the effects of antidepressants can be measured based on improved symptoms.
4. **Disgust** is an interesting feeling that is often associated with avoidance. This emotion that is associated with activation and connections between the left amygdala, the left inferior frontal cortex, and the insular cortex.
5. **Anger** is an important emotion that many people, adults and children alike, try to control. Anger is associated with activation of the right hippocampus, the amygdala, both sides of the prefrontal cortex and the insular cortex.
6. **Surprise** is an emotion that can either make you feel good or it can make you feel bad. Surprise activates the bilateral inferior frontal gyrus and the bilateral hippocampus. The hippocampus is strongly associated with memory, and the element of surprise is, by nature, associated with experiencing something that you do not remember or do not expect.

➤ **INVOLVEMENT OF AMYGDALA IN EMOTIONS:**

The amygdala is a small almond-shaped structure in the brain that plays a crucial role in processing emotions, especially fear and anxiety. It acts as a “threat detector” that helps us respond to potential dangers. The amygdala is responsible for recognizing emotional stimuli, processing fear and anxiety responses, associating emotions with memories, and influencing emotional behaviour and decision-making. It’s like an “emotional alarm system” that helps us react to threats and emotional situations, and is essential for our survival and well-being. The amygdala is a vital part of our brain’s emotional processing system, and damage to it can lead to difficulties in experiencing and managing emotions.

1. **Emotion Detection:** The amygdala is responsible for identifying emotional stimuli, such as threats, rewards, or social cues. It scans sensory input for emotional relevance, helping to determine whether a situation is dangerous, appealing, or neutral.
2. **Fear Processing:** The amygdala is particularly specialized for processing fear and anxiety. It helps associate stimuli with potential threats, triggering defensive responses like fight-or-flight reactions.
3. **Emotional Memory Formation:** The amygdala contributes to the formation of emotional memories, especially those related to fear or trauma. This helps learn from experiences and adapt behaviour to avoid similar situations in the future.
4. **Emotion Regulation:** The amygdala interacts with other brain regions, like the prefrontal cortex, to regulate emotional responses. It helps balance emotional intensity, preventing overwhelming or inappropriate reactions.
5. **Neurotransmitter Modulation:** The amygdala influences the release of neurotransmitters like dopamine, serotonin, and norepinephrine, which modulate emotional states, motivation, and arousal.
6. **Social Emotion Processing:** The amygdala is involved in processing social emotions like empathy, guilt, and shame. It helps recognize emotional cues from others, enabling social understanding and behaviour.

Damage to the amygdala can lead to:

- Impaired memory
- Social challenges
- Increased aggression
- Emotional instability
- Empathy difficulties
- Impulsive decisions

➤ INVOLVEMENT OF PREFRONTAL CORTEX IN EMOTIONS:

The PFC is like the “CEO of the brain”, managing many important functions. Damage to this area can impact daily life, relationships, and overall well-being. The prefrontal cortex acts as a “emotional manager,” balancing emotional responses with rational thinking, enabling us to navigate complex social situations and make informed decisions.

1. **Emotion Regulation:** The PFC helps manage and modulate emotional responses, preventing overwhelming feelings.
2. **Decision Making:** It considers emotional factors when making decisions, balancing rational thinking with emotional input.
3. **Emotional Expression:** The PFC influences how we express emotions, helping us choose appropriate responses.

4. **Empathy:** It enables us to understand others' emotions and perspectives.
5. **Fear Extinction:** The PFC helps overcome fear and anxiety by reprocessing emotional memories.
6. **Impulse Control:** It regulates impulsive behaviours driven by emotions.
7. **Mood Regulation:** The PFC contributes to maintaining a stable mood, mitigating extreme emotional shifts.

Damage to the prefrontal cortex (PFC) can affect:

- Self-control and decision-making
- Emotions and mood
- Planning and organization
- Memory and learning
- Personality and behaviour
- Social skills and empathy
- Movement and coordination
- Language and speech
- Problem-solving and multitasking

➤ **INVOLVEMENT OF INSULA IN EMOTIONS:**

The insula is a region in the brain that plays a vital role in various functions, including emotions, empathy, self-awareness, and interoception (the ability to sense one's own bodily sensations). It is often referred to as the "interoceptive cortex" and is responsible for helping us understand and sense our own emotions, hunger, thirst, pain, and other bodily sensations. The insula also helps us empathize with others, understand their perspectives, and recognize their emotions. It is a critical structure for our ability to experience and understand emotions, and is also involved in decision-making, impulse control, and cognitive processing. Damage to the insula has been linked to various disorders, including emotional processing deficits, empathy impairments, and interoceptive dysfunctions.

1. **Emotions:** The insula is involved in processing emotions, such as empathy, self-awareness, and emotional regulation.
2. **Interoception:** It helps us sense our body's internal state, including hunger, thirst, pain, and bodily sensations.
3. **Empathy:** The insula is activated when we observe others' emotions and physical sensations, enabling us to understand and share their feelings.
4. **Self-awareness:** It contributes to our sense of self and awareness of our own emotions, thoughts, and bodily sensations.

5. **Decision-making:** The insula is involved in decision-making processes, especially those related to emotional and social contexts.

6. **Sensory processing:** It processes sensory information from our body and environment, integrating it with emotional and cognitive information.

Damage to the insula has been linked to various disorders, including:

- Emotion regulation deficits
- Empathy impairments
- Interoception disorders (e.g., alexithymia)
- Anxiety and depression
- Chronic pain and fatigue syndromes

